Joint Control Barrier Function for Target Visibility and Thrust Deliverability

A camera-plane QP over the commanded inertial acceleration

Soft–Precise–Landing

31 August 2026

Scope. Visibility is fundamentally an *attitude* constraint (the body tilt shifts the field of view), but the attitude actually commanded is a byproduct of the desired inertial acceleration ${}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}$ an outer control loop produces. Rather than constraining a separately-defined tilt variable and then separately deciding how much vertical thrust to spend, this note poses one projection directly over ${}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}$ that jointly enforces (i) a field-of-view barrier condition and (ii) a thrust-deliverability bound, as a single constrained problem. The coupling between a commanded acceleration and the resulting feature motion is the **rotational part of the IBVS interaction matrix**, L_ω, evaluated at the measured image point — exact, *depth-independent*, no de-rotation or virtual frame required.

Notation

Symbol	Meaning
$\tilde{\boldsymbol{r}}_i,\ \tilde{r}_{i,k}$	measured camera-frame feature point i, tangent units, per axis k
$L_\omega(\tilde{\boldsymbol{r}}_i)$	rotational interaction matrix (roll/pitch columns), $\partial\dot{\tilde{\boldsymbol{r}}}_i/\partial\boldsymbol{\omega}_{\mathrm{rp}}$
$\boldsymbol{\omega}_{\mathrm{rp}} = [\omega_\phi, \omega_\theta]$	body roll/pitch rate (the *rotation-axis* vector)
$M = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$	lean→rotation map: a lean in direction $\boldsymbol{y}$ rotates about $M\boldsymbol{y}$
$\tilde{L}_{\omega,i} = L_\omega M$	effective coupling from acceleration-ratio direction to feature velocity
${}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}$	desired inertial *specific thrust* (thrust/mass), NED; this projection's decision varial
${}^{\mathcal{I}}a_{\mathrm{d},z}$	desired vertical acceleration component ($a_z \equiv \lvert {}^{\mathcal{I}}a_{\mathrm{d},z}\rvert$)
$\boldsymbol{y} = R_z(-\psi_{\mathrm{b}})({}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d},xy}/{}^{\mathcal{I}}a_{\mathrm{d},z})$	lateral-to-vertical acceleration ratio, expressed in image-axis coordinates
$\boldsymbol{y}_{\mathrm{curr}}$	the same ratio, evaluated at the current realized attitude
A_{cap}	vehicle's maximum achievable thrust-to-mass ratio
θ_{cap}	deliverable-tilt cap, $\lVert\boldsymbol{y}\rVert \le \tan\theta_{\mathrm{cap}}$
$\boldsymbol{d}_i$	exogenous (translation) feature drift, $\boldsymbol{d}_i = \dot{\tilde{\boldsymbol{r}}}_{i,\mathrm{obs}} - L_\omega\boldsymbol{\omega}_{\mathrm{rp}}^{\mathrm{curr}}$
$\mathcal{R}, f$	resolution, focal (px); $\boldsymbol{\varphi}_{\max} = \mathcal{R}/(2f)$ half-FoV (tangent), the FoV edge
τ, g	drift look-ahead; gravitational acceleration
κ_{az}	descent-rate relief gain (§**??**), acceleration per unit suppressed lean

1 Barrier on the real camera plane

Per image axis k, for feature point i,

$$\boxed{h_{i,k} = \varphi_{\max,k} - |\tilde{r}_{i,k}|}. \tag{1}$$

$h_{i,k} \geq 0 \iff$ feature point i is on the sensor on axis k. $\tilde{\boldsymbol{r}}_i$ is the *measured* camera-frame feature point — nothing is de-rotated.

2 Tilt→feature coupling, expressed in acceleration units

A camera rotation moves a feature point by the rotational part of the IBVS interaction matrix, a standard and **depth-independent** object evaluated at the measured point:

$$\dot{\tilde{\boldsymbol{r}}}_i = L_\omega(\tilde{\boldsymbol{r}}_i)\,\boldsymbol{\omega}_{\mathrm{rp}} + \boldsymbol{d}_i, \qquad L_\omega = \begin{bmatrix} xy & -(1+x^2) \\ 1+y^2 & -xy \end{bmatrix}_{(x,y)=\tilde{\boldsymbol{r}}_i}. \tag{2}$$

L_ω contains **no depth**. The drift is measured directly by stripping the rotational term from the observed feature velocity, $\boldsymbol{d}_i = \dot{\tilde{\boldsymbol{r}}}_{i,\mathrm{obs}} - L_\omega \boldsymbol{\omega}_{\mathrm{rp}}^{\mathrm{curr}}$ (using the true gyro rate), so it carries only genuine target motion, not our own tilting.

L_ω couples to a body *rotation rate* $\boldsymbol{\omega}_{\mathrm{rp}}$, but the quantity a lateral-acceleration command directly determines is not a rotation rate — it is the underactuated quadrotor's *lean*, the ratio of lateral to vertical thrust acceleration, ${}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d},xy}/{}^{\mathcal{I}}a_{\mathrm{d},z}$. Producing a steady lean in a given direction requires a rotation about the perpendicular axis; writing $\boldsymbol{y}$ for the acceleration ratio expressed in image-axis coordinates (a rotation by $R_z(-\psi_{\mathrm{b}})$ aligns the inertial x-y axes with the image axes),

$$\boldsymbol{y} = R_z(-\psi_{\mathrm{b}})\Big(\frac{{}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d},xy}}{{}^{\mathcal{I}}a_{\mathrm{d},z}}\Big), \qquad \boldsymbol{\omega}_{\mathrm{rp}} = M\boldsymbol{y}, \quad M = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}. \tag{3}$$

$\boldsymbol{y}$ carries no information beyond the acceleration ratio itself — it is not a separate physical variable, only that ratio written in the coordinate frame L_ω is defined in. Substituting (**??**) into (**??**) gives the feature velocity directly as a function of the acceleration command:

$$\tilde{L}_{\omega,i}\boldsymbol{y} = \underbrace{\Big(\frac{\tilde{L}_{\omega,i}\,R_z(-\psi_{\mathrm{b}})}{a_z}\Big)}_{N_i(a_z)}\,{}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d},xy}, \qquad \tilde{L}_{\omega,i} = L_\omega M. \tag{4}$$

$N_i(a_z) = \tilde{L}_{\omega,i}R_z(-\psi_{\mathrm{b}})/a_z$ is the barrier's Jacobian with respect to the lateral acceleration directly. It depends on a_z — the same quantity the joint projection below also solves for — which is exactly the coupling that projection is built to resolve, rather than treating a_z as an external given. For a finite acceleration change the exact map is the rotation homography of the measured point; the first-order form

$$\tilde{\boldsymbol{r}}_i({}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d},xy}) \approx \tilde{\boldsymbol{r}}_i + N_i(a_z)\,{}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d},xy} - \tilde{L}_{\omega,i}\boldsymbol{y}_{\mathrm{curr}} \tag{5}$$

is the local linearization, accurate over one control step. $\boldsymbol{y}_{\mathrm{curr}}$ is read directly from the realized attitude (not from ${}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}$), so it needs no rotation from acceleration units.

3 The visibility–deliverability projection

Two physical constraints bind the same variable ${}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}$: the barrier condition of (**??**) along the feature response (**??**)–(**??**), and the requirement that the commanded thrust vector be one the vehicle can actually deliver. Writing ${}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}^{\star}$ for the conditioned output,

$$\boxed{\begin{aligned} &{}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}^{\star} = \arg\min_{{}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}} \|{}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}} - {}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}^{0}\|^2 \\ &\text{s.t.}\quad \left|\tilde{\boldsymbol{r}}_i + N_i(a_z)\,{}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d},xy} - \tilde{L}_{\omega,i}\boldsymbol{y}_{\mathrm{curr}} + \tau\boldsymbol{d}_i\right| \preceq \boldsymbol{\varphi}_{\max}, \qquad \left\|{}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}\right\| \le A_{\mathrm{cap}} \end{aligned}} \tag{6}$$

where ${}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}^{0}$ is the unconstrained desired acceleration, the target this projection stays closest to. The first constraint is the per-feature FoV box, written directly on ${}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d},xy}$; the second is the thrust-deliverability sphere. Because ${}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}$ is already the specific thrust, the vehicle's thrust vector is $m\,{}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}$ and the bound is simply $\|{}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}\| \le A_{\mathrm{cap}}$; no gravity shift enters. *Correction, 2026-09-03:* earlier revisions of this note wrote this constraint as $\|{}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}} + g\boldsymbol{e}_3\| \le A_{\mathrm{cap}}$ and called it the "true" deliverability sphere. That expression is the *vehicle's* acceleration, which is $\mathbf{0}$ at hover, so it bounded deviation from hover rather than thrust — at zero lateral demand it admitted $2.39\,g$ on a vehicle capable of $1.389\,g$. The implementation inherited the error from this note (in two places) and is corrected behind `CBF_SPHERE_TRUE_THRUST`, default on. Both are convex for a fixed a_z; since $N_i(a_z)$ itself depends on a_z, the two constraints are resolved jointly rather than sequentially: the lateral acceleration satisfying the box and the vertical acceleration satisfying the sphere are solved for together, not one after the other. ${}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}^{\star}$ then defines the desired attitude and thrust magnitude for the inner attitude-tracking loop.

- **Directional** — only the outward-binding box row is active; inward (recovery) and tangential accelerations are free. No strangling.
- **Envelope-aware** — the deliverability side of (**??**) is the vehicle's actual maximum thrust-to-mass ratio A_{cap}, not a fixed angular cap applied *as if* a_z were fixed at hover; it tightens or loosens correctly as a_z moves away from hover.
- **Forward-invariant** — the look-ahead shifts the box anchor by $\tau\boldsymbol{d}_i$, so the projection accounts for centroid drift over the next τ s.
- **Bounded by construction** — because ${}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}^{\star}$ always satisfies the sphere constraint, the resulting acceleration ratio $\boldsymbol{y}^{\star} = R_z(-\psi_{\mathrm{b}})\,{}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d},xy}^{\star}/a_z^{\star}$ satisfies $\|\boldsymbol{y}^{\star}\| \le \tan\left(\arccos(a_z^{\star}/A_{\mathrm{cap}})\right)$ at the solved $a_z^{\star}$ automatically. Note this is an *a_z-dependent* bound: it equals $\tan\theta_{\mathrm{cap}}$ only at $a_z^{\star} = g$, and is looser below hover / tighter above it ($a_z = 6.0$: 2.04 vs 0.96; $a_z = 12.0$: 0.54). An earlier revision claimed a fixed $\tan\theta_{\mathrm{cap}}$ came free here, which is why the implementation needed a post-projection clip on the derived ratio; that clip computes exactly $\arccos(a_z/A_{\mathrm{cap}})$ and is the principled bound, not a workaround.
- **Descent-rate aware** (§**??**) — while the box actively suppresses lateral authority, the vertical component is relaxed toward hover, in proportion to that suppression and clamped so it can only *slow* a descent, never reverse it; the trade lives inside (**??**) so the output stays box- and sphere-consistent.

3.1 Descent-rate relief under an active box

When the FoV box is binding it removes lateral acceleration the outer loop asked for: the realized lean $\boldsymbol{y}^\star$ falls short of the desired $\boldsymbol{y}^0 = R_z(-\psi_\mathrm{b})({}^{\mathcal{I}}\boldsymbol{a}^0_{\mathrm{d},xy}/{}^{\mathcal{I}}a^0_{\mathrm{d},z})$ by

$$\Delta_\theta \;=\; \left\|\boldsymbol{y}^0 - \boldsymbol{y}^\star\right\|, \tag{7}$$

the magnitude of lean the box could not grant this step. Pressing on at the full commanded closing rate while $\Delta_\theta > 0$ spends the remaining descent budget without the lateral error being corrected — the marker leaves the FoV, or touchdown arrives, before convergence. The projection therefore spends part of that budget on time instead: the vertical component is pulled toward the hover value in proportion to Δ_θ,

$$\boxed{{}^{\mathcal{I}}a^\star_{\mathrm{d},z} \;=\; \min\Big({}^{\mathcal{I}}a_{\mathrm{d},z},\ \max\left({}^{\mathcal{I}}a^0_{\mathrm{d},z} - \kappa_{az}\,\Delta_\theta,\ -g\right)\Big) \quad \text{when } {}^{\mathcal{I}}a^0_{\mathrm{d},z} > -g} \tag{8}$$

and left unchanged otherwise. The interpretation is exact per axis of (**??**): an A_cap budget spent laterally is not available vertically and vice versa; (**??**) makes the vertical side of that trade explicit whenever the box has forced the lateral side. Two bounds make it safe as a one-way relief:

- the inner $\max(\cdot, -g)$ clamps the result at the hover acceleration $-g$, so this term alone can bring a commanded descent toward zero closing rate but *never* past it into a commanded climb;
- the outer $\min({}^{\mathcal{I}}a_{\mathrm{d},z}, \cdot)$ keeps any lift the deliverability sphere or the outer loop already commanded (${}^{\mathcal{I}}a_{\mathrm{d},z} < -g$): the relief only ever adds toward-hover, it never removes lift.

The relief is recomputed from the unconstrained ${}^{\mathcal{I}}a^0_{\mathrm{d},z}$ each outer iterate, so it does not accumulate across iterates, and it re-enters $N_i(a_z)$ on the next iterate — the lateral box and the vertical relief converge together, not one after the other. $\kappa_{az} = 0$ disables it and recovers (**??**) unchanged.

4 Feasibility

The box constraint alone is **infeasible** iff $\varphi_{\max,k} < 0$ for some k (the marker subtends the full FoV). The sphere constraint is always feasible by construction (it is satisfied by ${}^{\mathcal{I}}\boldsymbol{a}_\mathrm{d} = \boldsymbol{0}$). Genuine joint infeasibility — the box demanding more lateral acceleration than A_cap can deliver at any achievable a_z — reduces to

$$\sigma_{\min}(\tilde{L}_{\omega,i})\,(\tan\theta_\mathrm{cap} - \|\boldsymbol{y}_\mathrm{curr}\|) \;\geq\; \max_{i,k}\left(|\tilde{r}_{i,k}| - \varphi_{\max,k}\right)_+ + \bar{d} \quad \forall i, \tag{9}$$

evaluated at the solved $a_z^\star$ rather than at a fixed hover-assumed θ_cap: joint feasibility holds whenever some achievable a_z satisfies (**??**), a weaker requirement than demanding it hold at the hover value of a_z alone.

Open, flagged 2026-09-03. (**??**) is written purely in tilt terms. With the deliverability constraint corrected to $\|{}^{\mathcal{I}}\boldsymbol{a}_\mathrm{d}\| \leq A_\mathrm{cap}$ *and* a_z a solved variable, a tilt-only inequality is no longer a sufficient condition on its own: it needs a thrust conjunct, and the substitution $\theta_\mathrm{cap} \to \arccos(a_z^\star/A_\mathrm{cap})$ alone has not been shown to close it. A second coupling is unaddressed — the descent-rate relief of §**??** shrinks $|a_z|$, which *inflates* $N_i(a_z)$ and so tightens the box in ${}^{\mathcal{I}}\boldsymbol{a}_\mathrm{d}$ units, meaning relief and box can fight across outer iterates. The solver also runs a fixed iterate budget with no convergence test and no residual logged, so convergence is currently unobserved either way. Deriving the corrected condition is deliberately left open here rather than guessed at.

5 Assumptions

1. **The coupling $\tilde{L}_{\omega,i} = L_\omega M$ and its sign convention are calibrated.** The fixed 90° map $\boldsymbol{\omega}_{\mathrm{rp}} = M\boldsymbol{y}$ bridges the acceleration-ratio direction and the rotation-axis rate that L_ω couples to.

2. **The conditioned acceleration is realized**: the inner attitude loop tracks the attitude built from ${}^{\mathcal{I}}\boldsymbol{a}_{\mathrm{d}}^{\star}$ on a timescale fast relative to the outer loop.

3. **The drift $\boldsymbol{d}_i$ is the translation part** of the observed feature velocity, $\dot{\tilde{\boldsymbol{r}}}_{i,\mathrm{obs}} - L_\omega \boldsymbol{\omega}^{\mathrm{curr}}$, entering as *measured* motion — no separate disturbance term is introduced.

4. **First-order L_ω over the cycle.** Exact is the rotation homography of $\tilde{\boldsymbol{r}}_i$; the linearization (**??**) is accurate over one control step.

5. $A_{\mathbf{cap}}$ **is the vehicle's true achievable thrust-to-mass ratio**, used directly rather than a fixed angular margin computed from it at hover.

6 Scale-freeness and depth-freeness

The barrier (**??**), the coupling (**??**), and the projection (**??**) contain only the measured image point $\tilde{\boldsymbol{r}}_i$, $\boldsymbol{\varphi}_{\max} = \mathcal{R}/(2f)$, the body rate, the commanded acceleration, and the vehicle's own A_{cap} and g. L_ω **is depth-independent by construction**, and A_{cap}, g are fixed vehicle/physical constants, not measurements of target depth or altitude — the whole formulation remains manifestly scale- and depth-free.